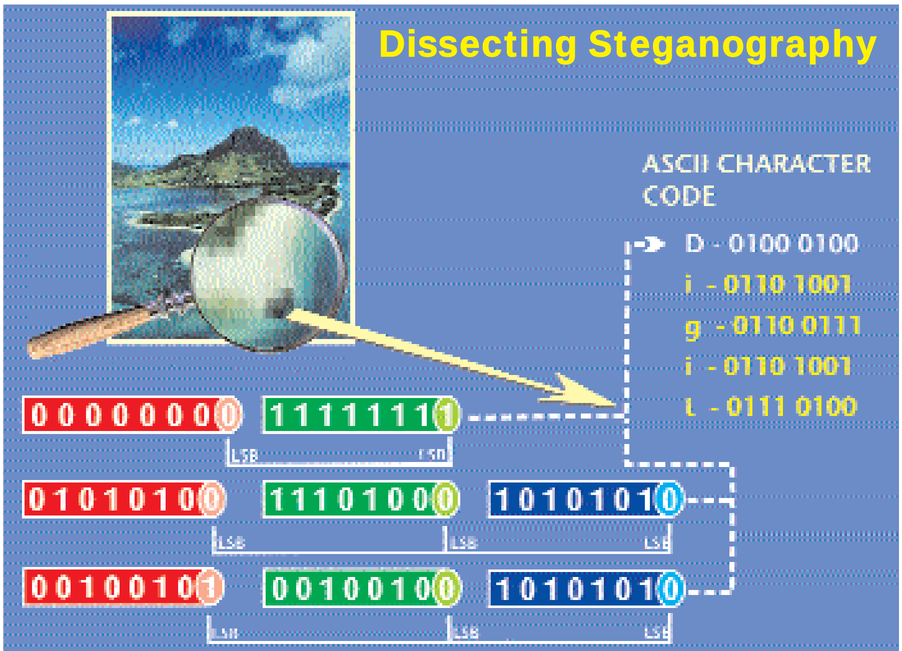

Steganography is the term used for hiding messages within an image. Any colour pixel is made of a combination of Red-Green-Blue mode (RGB) wherein each RGB component consists of 8 bits.

If the letters in ASCII are to be represented within the colour pixels, the rightmost digit, called the Least Significant Bit (LSB), can be altered. Any variation in the value of this bit leads to a very minimal variation in colour.

If we have to hide the word 'Digit' in the image, we take the LSB of every colour and hide each bit of the word in its RGB combination. To insert the letter 'D', we modify three colour pixels with 3 bits in each colour pixel, we utilise 14 colour pixels to hide the entire word with only 1 bit in the 14th pixel.

within still images, audio or even video. Steganography software store the classified information in the least significant bits of a digitised file, bits that can be changed in subtle ways that cannot be detected by the human eye or ear.

The information is first encrypted as a data file. Once you have this, you need to select a carrier file on which the data file can be encoded. The carrier file can be any graphic or audio file eight times the size of the data file. For example, one could hide information in, say, the nose of a puppy on a pet-adoption Web site. The intended recipients can then extract the information.

Message in a bottle

There are different kinds of carriers, which can carry your message across mediums. The earliest form of steganography was hiding messages in text. With the evolution of multimedia applications, it became easier to hide messages in still images since these are larger files. Audio and video carrier files are less perceptible, hence providing higher invisibility. The technique of steganography is more or less the same in each carrier file.

Messages in text: Secret messages can be hidden in text format by reframing the text of the carrier file, while maintaining the context. One form of Steganography is a program called Spam Mimic. Based on a set of rules called a mimic engine by Peter Wayner, it encodes your message into what looks just like your typical, quickly deleted spam message. However, hiding a message in plain text is a thing of the past, as people are suspicious of irrelevant text.

Messages in still images: This is the most popular method of steganography as minor changes in colour are unnoticeable to the human eye. Outguess, a universal steganography tool, comprises data-specific handlers (code) that extract redundant bits in an image and write them back after modification. Outguess can use any kind of data file as a carrier, as long as a handler is provided. The amount of message that can be encrypted depends upon the size of the carrier.

Messages in audio: Messages in audio are always sent along with ambient noise. The data is hidden in the heart of the Layer III encoding process of an MP3 file, namely, the inner loop during compression. The inner loop limits the input data and increases the step size until the data can be coded with the available number of bits.

Stegano Tools

Various tools are available for concealing and sending messages using steganography. Some of the commonly used tools are:

- Steganos Security Suite 4 uses powerful 128-bit encryption. It would take 1 billion powerful computers millions of years to try every combination to gain access to your personal information. This software uses steganography along with encryption to completely secure your data.
  Web site: www.steganos.com

- Datamark Technologies has a number of products lined up for steganography. They currently market four digital steganography products, namely, StegComm for confidential multimedia communication, StegMark for digital watermarking storage media, StegSafe for digital storage and linkage, and StegSign for e-commerce transactions.
  Web site: http://www.datamark-tech.com/datamark.html
- Blindside is an application of steganography that allows you to conceal a single file or set of files within a standard computer image. The new image looks identical to the original, but can contain up to 50 k of data. The hidden files can also be password encrypted, to prevent unauthorised access.
  Web site: http://www.blindside.co.uk
- MP3Stego hides information in MP3 files during the compression process. The data is first compressed, encrypted and then hidden in the MP3 bit stream. Although MP3Stego was written with steganographic applications in mind, it can also be used as a copyright marking system for MP3 files.
  Web site: http://www.cl.cam.ac.uk/~fapp2/steganography/mp3stego